

Quentin FRUYTIER

Ph.D. Candidate

UT Austin

M.Sc. & B.Sc. Graduate

McGill University

✉ qfruytier@gmail.com

☎ +1 (737) 351-4678

🌐 in/quentin-fruytier/

🌐 [Personal Website](#)

Ph.D. Supervisors

[Aryan Mokhtari](#)

[Sujay Sanghavi](#)

M.Sc. Supervisors

[Abbas Khalili](#)

[Tim Hoheisel](#)

Patents

Latent Distribution Shaping:

App No. 19/337202 2026

Programming Skills

Python (PyTorch & TensorFlow),
C, C++, Matlab, and R.

Languages

Bilingual French (native) and
English

Nationalities

United States, Canada, Belgium

Domain Knowledge

Optimization & Theory: EM, Mirror
Descent, SGD, non-convex
convergence analysis.

Statistical ML: mixture models, kernel
methods (MMD), time-series
forecasting, disentanglement.

Architectures: Transformers / ViT,
Autoencoders, Mixtures of Experts,
Diffusion (continuous & discrete) .

Education

UT Austin • Ph.D. in Electrical Engineering • 4.0/4.0 GPA

Expected 2028

McGill University • Masters in Mathematics • 4.0/4.0 GPA

August 2023

McGill University • Joint Honours Math & Comp. Sci. • 3.8/4.0 GPA

June 2021

Publications

[Sculpting the Latent Space With MMD \(accepted at SDM 2026\)](#)

2026

Introduced the Latent Predictability Score, an unsupervised lower bound on latent entanglement, showing VAE- and adversarial-based baselines fail to learn statistically independent representations. Used Maximum Mean Discrepancy to sculpt the aggregate posterior toward arbitrary target priors — **70% reduction in detected entanglement** over the leading baseline with no reconstruction trade-off.

[Learning Mixtures of Experts with EM \(accepted at ICML 2025\)](#)

2025

Proved a novel equivalence between EM and Projected Mirror Descent for Mixtures of Experts, establishing matching local linear convergence rates for both EM and SGD; in small-scale experiments, EM reached a better optimum in fewer iterations than SGD

Projects

[Quantitative ETF Forecasting App](#)

2026

A self-hosted, public web app that manages capital allocation based on ETF forecast behavior using Amazon's Chronos-2 time-series model alongside classical signals (SMA-50, momentum, realized volatility, M2 money supply, inflation); reproducible statistical analysis, code fully open.

[Statistical Ranking for n-vs-n Games](#)

2024

Elo/Glicko-style rating algorithms recovering latent skill from noisy pairwise outcomes.

Work Experience

Machine Learning Research Intern

Summer 2025

InterDigital Communications AI Lab

Los Altos, CA

Developed a Maximum Mean Discrepancy framework to shape neural latent spaces to arbitrary target distributions — independence, semantic alignment, prescribed geometry — with provable guarantees; filed as U.S. patent *Latent Distribution Shaping* and accepted at SDM 2026. Applied to **wireless CSI representation learning and cross-model interoperability**.

Data Science Intern at Nectar

Summer 2021

Nectar Technologies Data Science

Montreal, QC

RNN and AE training on time series sensor data collected from within beehives.

Software Engineer Intern at Matrox/Akamai

Summers 2019/2020

Akamai Technologies Image Manager

Montreal, QC/Ottawa, ON

Teaching Experience

Graduate Teaching Assistant

2026

Graduate topics course on Generative Models in Machine Learning

UT Austin Electrical and Computer Engineering Department

Austin, TX

Teaching Assistant

2019-2022

Principles of Statistics 1, Calculus 2, Foundations of Programming (Python/ Java)

McGill University Department of Mathematics and Statistics

Montreal, QC

Awards

ICML 2026 Silver Reviewer Award

May 2026

Basdall Gardner Memorial Graduate Fellowship in Engineering

2023 - 2026

UT Austin fellowship for academic excellence

Graduate Excellence Fellowship Award

May 2022

McGill University award for academic excellence